IAD-Risk: Risk-Aware Identity-Agenda Disentanglement for Scholarly Work Deduplication

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Abstract

Scholarly entity matching must distinguish whether two records describe the same work, not merely whether they discuss the same research agenda. Modern scientific representations and pair classifiers often improve semantic matching, but a high semantic score can also connect different papers in the same task, benchmark, or citation community. This paper studies this failure mode as *identity-agenda confusion* and proposes IAD-Risk, a risk-aware framework that separates identity evidence from agenda evidence before making merge decisions. IAD-Risk combines stratified data provenance, identity and agenda relation heads, and false-merge risk gating to prevent agenda-level hard negatives from entering automatic merge clusters. We evaluate the framework with IAD-Bench, a benchmark contract that separates DeepMatcher gold labels, SciRepEval/SciDocs-style proxy labels, and OpenAlex/OpenCitations silver hard negatives. In the Open-v2 evidence snapshot, single-space scientific representation baselines show HNFMR 0.790–0.999 on the full pair scope, whereas transformer-backed IAD-Risk variants report same-work F1=0.980 and HNFMR=0.000 on the held-out test scope. The results support a conservative conclusion: identity-agenda disentanglement is a practical mechanism for safer scholarly deduplication, while additional manual validation and broader source-heldout validation remain necessary before making broad method-ranking claims.

Keywords: scholarly entity matching, work deduplication, identity-agenda disentanglement, false-merge risk, provenance-aware evaluation, scientific document representation

1. Introduction

Scholarly knowledge systems rely on entity matching to decide whether multiple records describe the same scientific work. Digital libraries, literature review tools, citation graphs, recommender systems, and evidence synthesis pipelines all require stable work identities because a false merge can combine evidence from different papers and distort downstream analysis. This paper focuses on scholarly work deduplication: given two scholarly records with titles, abstracts, authors, venues, identifiers, topics, and citations, the system must decide whether the records can be safely merged as the same work.

The central challenge is that topical similarity is not identity evidence. Two papers can share a task, dataset, method family, citation neighborhood, or OpenAlex topic while making different contributions. Scientific representation models such as SPECTER and SciNCL are valuable because they bring related papers close in embedding space [1, 2], but the same behavior can create false-merge risk when a deduplication system treats semantic closeness as work identity. This failure is particularly important in scholarly settings because false merges are harder to repair than missed duplicates: once distinct papers are clustered as one work, citation counts, recommendation signals, and review evidence can be contaminated.

We formulate this failure mode as identity-agenda confusion. Identity refers to whether two records denote the same scholarly work. Agenda refers to whether two records discuss the same research problem, benchmark, method family, or citation community. Existing entity matching methods [3–5] typically optimize pair classification, while scholarly document representation methods optimize semantic relatedness. Neither objective alone forces the model to learn that same-agenda pairs can be high-risk non-identities.

We propose IAD-Risk, a risk-aware identity-agenda disentanglement framework for scholarly work deduplication. IAD-Risk learns separate relation signals for same-work identity, same-agenda relatedness, and agenda-non-identity hard negatives, then derives a false-merge risk score from these signals. A pair is eligible for automatic merging only when identity evidence is sufficiently strong and agenda-non-identity risk remains below a merge threshold. This selective decision rule turns deduplication from a high-recall similarity problem into a risk-calibrated safety problem.

The paper makes three contributions. First, it defines identity-agenda confusion as a targeted failure mode in scholarly entity matching and introduces hard-negative false-merge rate as a safety metric. Second, it presents

IAD-Bench, a provenance-aware benchmark contract that keeps gold, proxy, silver, and human-reviewed labels separate. Third, it evaluates IAD-Risk under the same pair contract used by representation and entity-matching baselines, while keeping clustering and artifact-audit interfaces as reproducible extensions rather than unsupported primary evidence. The current evidence supports targeted false-merge suppression under stratified evaluation; it does not support a broad method-ranking claim or a claim that silver labels replace human gold.

Table 1: Contribution-evidence summary. The table states what each contribution is, which evidence supports it, and which stronger claim remains outside the current manuscript.

Contribution	Supporting evidence in this paper	Claim boundary
Identity-agenda confusion as a scholarly deduplication failure mode.	Problem formulation, hard-negative false-merge rate, and Open-v2 hard-negative evidence.	Identifies a concrete safety failure mode, not a prevalence estimate for all scholarly corpora.
IAD-Bench as a provenance-aware pair contract.	Gold, proxy, silver, and manual-validation layers with label source, label strength, provenance, split, and hard-negative fields.	Defines an evaluation contract; proxy and silver labels are not human identity gold.
IAD-Risk as a risk-aware merge mechanism.	Open-v2 evidence snapshot with single-space baseline HNFMR 0.790–0.999 and held-out IAD-Risk same-work F1=0.980, HNFMR=0.000.	Supports targeted false-merge suppression, not a broad method-ranking claim.

2. Related Work

Entity resolution research defines the broader pipeline in which scholarly deduplication is situated. Classical record linkage formalizes the decision of whether two records refer to the same entity [3], while modern surveys describe end-to-end entity resolution as a workflow involving blocking, matching, and clustering under scale and heterogeneity constraints [6]. System work such as Magellan further emphasizes that practical entity matching requires sampling, labeling, debugging, workflow management, and production execution rather than only a matching function [7]. IAD-Risk follows this pipeline view, but focuses on a specific safety gap in scholarly data: a pair can be highly related in topic or citation context while still being an unsafe merge.

Neural entity matching methods provide the closest algorithmic baselines for learning pair-level match decisions. DeepER represents tuples with neural distributed representations and supports blocking over learned tuple vectors [8]. DeepMatcher explores design choices for neural entity matching models [4], and Ditto shows that transformer-based matching can benefit from pretraining, fine-tuning, and data augmentation [5]. RoBERTa-style pair encoders [9] are therefore natural baselines for this paper. The distinction is that these methods usually optimize a same-entity classification objective, whereas IAD-Risk adds explicit same-agenda and agenda-non-identity heads so that semantic relatedness can become blocking evidence rather than merge evidence.

Scientific document representation methods are strong baselines because they intentionally capture citation, topical, and semantic relatedness. SPECTER uses citation-informed supervision to learn document-level scientific representations and introduced SciDocs as a document-level evaluation benchmark [1]. SciNCL improves scientific representations with neighborhood contrastive learning [2], and SciRepEval broadens scientific representation evaluation across classification, regression, ranking, and search while releasing the SPECTER2 model family [10]. These methods are useful for candidate generation and relatedness scoring, but their optimization target is not conservative work identity. IAD-Risk therefore treats high scientific relatedness as a possible false-merge risk unless identity evidence is also strong.

Scientific language resources and open scholarly metadata make reproducible benchmark construction possible, but they do not remove the need for provenance-aware evaluation. SciBERT shows that domain-specific pretraining on scientific text improves scientific NLP tasks [11], and S2ORC provides metadata, abstracts, references, and structured full text for many open-access papers [12]. OpenAlex provides a large open index of scholarly works and concepts [13], while OpenCitations provides open citation data [14]. IAD-Bench uses these resources to construct same-work, same-agenda, and agenda-non-identity evidence, but separates gold, proxy, and silver labels so that citation or topic proximity is never presented as human identity gold.

Table 2: Positioning against the closest lines of work. IAD-Risk differs by treating agenda relatedness as a risk signal for merge decisions rather than as direct identity evidence.

Line of work	Primary optimization target	Limitation for scholarly deduplication	IAD-Risk distinction
End-to-end entity resolution systems	Blocking, matching, labeling, and deployment workflows.	Workflow abstractions do not separate topical relatedness from work identity.	Adds identity-agenda semantics and false-merge risk gates inside matching.
Neural entity matching	Pair-level same-entity classification from tuple or text encoders.	Strong text similarity can still link different papers in one agenda.	Trains same-work, same-agenda, and agenda-non-identity heads with provenance-aware masking.
Scientific document representations	Citation-informed or contrastive scientific relatedness embeddings.	Relatedness embeddings support retrieval but are unsafe as direct merge evidence.	Uses representation similarity as candidate and agenda evidence, then blocks risky merges.
Open scholarly metadata benchmarks	Public metadata, citation, and topic signals for scalable evaluation.	Topic or citation proximity is silver evidence, not human identity gold.	Reports gold, proxy, and silver strata separately and ties claims to label strength.

3. Problem Formulation

Let d_i and d_j be two scholarly records. The target identity label y^I_{ij} indicates whether the records describe the same scholarly work. The agenda label y^A_{ij} indicates whether the records discuss a related research agenda. A hard negative is a pair with high agenda evidence but negative identity evidence:

$$y^A_{ij} = 1, \quad y^I_{ij} = 0.$$

This pair should not be merged even if the documents are semantically close.

We evaluate merge safety with hard-negative false-merge rate:

$$\text{HNFMR} = \frac{\#\{(i,j) : \hat{m}_{ij} = 1, y^A_{ij} = 1, y^I_{ij} = 0\}}{\#\{(i,j) : y^A_{ij} = 1, y^I_{ij} = 0\}},$$

where $\hat{m}_{ij}$ is the automatic merge decision. The goal is not only to maximize same-work F1, but also to control HNFMR under explicit evidence provenance.

3.1. Notation and Relation Semantics

Table 3 summarizes the notation used throughout the paper. The key distinction is that identity variables describe whether two records denote the same work, while agenda variables describe topical or citation-level relatedness. This distinction prevents a high relatedness score from being interpreted as direct merge evidence.

Table 3: Notation and relation semantics. The paper uses separate symbols for identity, agenda relatedness, agenda-non-identity risk, and merge decisions.

Symbol or field	Meaning	Role in evaluation
d_i, d_j	Two scholarly records.	Input pair for scoring and merge decisions.
y_{ij}^{I}	Identity label: the records describe the same work.	Gold identity evidence when supported by the label source.
y_{ij}^{A}	Agenda label: the records share a research agenda.	Proxy or silver relatedness evidence, not sufficient for merging.
y_{ij}^{ANI}	Agenda-non-identity label.	Hard-negative supervision for pairs that are related but should not merge.
p_{work}	Predicted same-work probability.	Positive merge evidence.
p_{agenda}	Predicted agenda-relatedness probability.	Relatedness evidence that can increase risk when identity is weak.
p_{ani}	Predicted agenda-non-identity probability.	Direct risk evidence against automatic merging.
p_{risk}	Derived false-merge risk score.	Gate for rejecting or deferring risky pairs.
$\hat{m}_{ij}$	Automatic merge decision.	Determines FMR and HNFMR.
label strength	Gold, proxy, silver, or manual validation layer.	Controls how a row can be used in training, evaluation, and claims.

4. IAD-Bench

IAD-Bench is a benchmark contract for organizing scholarly pair evidence. It is not a claim that every label has the same evidential status. Each pair stores relation labels, expected identity labels, expected agenda labels, label source, label strength, provenance, split, and hard-negative level.

Table 4: IAD-Bench label layers. Gold, proxy, and silver labels are reported separately to avoid overstating evidence strength.

Layer	Source	Use in this paper
Gold	DeepMatcher / py_entitymatching	Same-work identity evaluation
Proxy	SciRepEval / SciDocs-style proximity	Same-agenda analysis
Silver	OpenAlex / OpenCitations	Agenda-non-identity hard negatives
Manual validation	Independent human review	External validation target

The Open-v2 benchmark variant used in the core evidence combines 415 DeepMatcher gold pairs with 10,000 OpenAlex-derived silver hard negatives, producing 10,415 pair records over 1,737 documents. The DeepMatcher component provides same-work gold labels, while the OpenAlex component stresses same-agenda non-identity behavior. This design is intentionally conservative: OpenAlex and OpenCitations are used to expose false-merge risk, not to replace human gold.

Table 5: Open-v2 benchmark composition used in the current evidence snapshot. The table separates identity evidence from hard-negative stress evidence before any metric is reported.

Evidence stratum	Pair count	Evaluation role	Claim boundary
DeepMatcher gold identity	415	Measures same-work matching ability.	Supports identity evaluation within the imported gold benchmark pairs.
OpenAlex and OpenCitations silver hard negatives	10,000	Stresses agenda-level false-merge behavior.	Supports hard-negative safety analysis, not human non-identity gold.
Open-v2 combined pair scope	10,415	Provides the shared pair contract for the reported evidence snapshot over 1,737 documents.	Supports the Open-v2 mechanism demonstration; broader source-heldout claims require additional artifacts.

5. Method

5.1. Overview

IAD-Risk converts scholarly deduplication into selective risk-aware matching. Given a candidate pair, the model estimates identity, agenda, and agenda-non-identity signals, then derives a false-merge risk score. The final merge decision is made only when identity evidence exceeds a same-work threshold and risk evidence remains below blocking thresholds. Figure 1 summarizes the workflow.

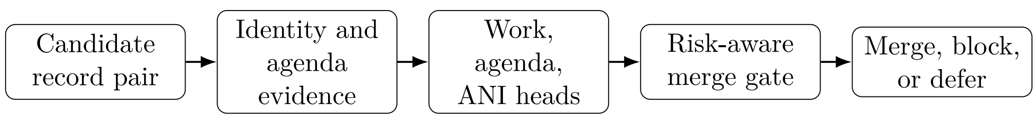

Figure 1: IAD-Risk pipeline. The framework first separates identity evidence from agenda evidence, then predicts relation signals and derives a false-merge risk score before applying a conservative merge gate.

5.2. Identity and Agenda Signals

IAD-Risk separates the representation space into identity-oriented and agenda-oriented evidence. Identity evidence includes strong identifiers, near-duplicate titles, author overlap, year and venue agreement, and work-level metadata. Agenda evidence includes shared research topics, similar abstracts, citation neighborhoods, shared references, and scientific representation similarity. The separation is necessary because the same feature can have different meanings: high embedding similarity may indicate same work, but it may also indicate two different papers in the same research area.

5.3. Multi-Task Relation Heads

The model predicts three pair-level relation quantities: p_{work}, p_{agenda}, and p_{ani}. The same-work head estimates identity. The same-agenda head estimates topical relatedness. The agenda-non-identity head detects pairs that are related but should not be merged. The false-merge risk score is then derived from these heads:

$$p_{\text{risk}} = \max\{p_{\text{ani}},\ p_{\text{agenda}}(1 - p_{\text{work}})\}.$$

This construction makes the safety signal high when a pair is explicitly predicted as agenda-non-identity, or when it is agenda-related but lacks strong identity evidence.

5.4. Training Objective

The learning objective combines identity, agenda, and agenda-non-identity supervision without treating every source as the same label type; the false-merge risk score is derived after the relation heads are estimated. For a pair (i, j), the model minimizes

$$\mathcal{L}_{ij} = \lambda_w \mathrm{CE}(y_{ij}^{I}, p_{\mathrm{work}}) + \lambda_a \mathrm{CE}(y_{ij}^{A}, p_{\mathrm{agenda}}) + \lambda_n \mathrm{CE}(y_{ij}^{ANI}, p_{\mathrm{ani}}),$$

where CE is binary cross-entropy and $\lambda_w, \lambda_a, \lambda_n$ control the contribution of each supervision channel. The key design choice is provenance-aware masking: a pair contributes only to the heads for which its label source is valid. For example, DeepMatcher pairs supervise same-work identity, whereas OpenAlex/OpenCitations silver hard negatives supervise agenda-non-identity risk without being promoted to human gold identity labels.

5.5. Risk-Aware Merge Decision

The merge policy is:

$$\hat{m}_{ij} = 1 \quad \text{iff} \quad p_{\mathrm{work}} \geq \tau_w, \quad p_{\mathrm{ani}} < \tau_a, \quad p_{\mathrm{risk}} < \tau_r, \quad \text{cannot\_link}_{ij} = 0.$$

This rule is deliberately selective. It can reject or defer high-risk pairs instead of forcing every semantically related pair into a binary match decision. The same decision interface also supports constrained union-find clustering, where cannot-link evidence prevents transitive false merges.

5.6. Failure-Control Rationale

IAD-Risk is designed as a failure-control framework rather than as another similarity scorer. Table 6 summarizes how each design choice addresses a specific false-merge pathway. The table also states the remaining boundary for each control, because a safety mechanism should make its unsupported cases explicit.

Table 6: Failure-control rationale of IAD-Risk. Each component targets a concrete way in which agenda-level similarity can be mistaken for work identity.

Failure pathway	Design response	Remaining boundary
Topically close papers receive high semantic similarity.	Separate agenda evidence from identity evidence before the merge gate.	Requires the benchmark to include agenda-level hard negatives.
Silver metadata is treated as if it were human gold.	Preserve label strength and provenance, and mask supervision by valid label source.	Silver labels support stress testing, not exhaustive truth claims.
Pairwise errors contaminate clusters through transitivity.	Apply risk gating before constrained union-find and respect cannot-link evidence.	Cluster-level guarantees require complete cannot-link coverage.
Thresholds turn a classifier into an unsafe automatic merger.	Expose work, ANI, and risk thresholds and evaluate held-out merge decisions.	Threshold transfer should be rechecked under new source distributions.
Proxy labels are over-interpreted.	Keep proxy rows separate unless the label source supports the relation being evaluated.	Proxy rows remain non-human evidence even when reproducible.

5.7. Implementation Details

The reported transformer variants use frozen scientific document encoders rather than end-to-end fine-tuning. For each candidate pair, the implementation builds three feature groups: identity features combine transformer distances with identifiers, author overlap, venue, and year signals; agenda features combine semantic similarity with topic and citation overlap; agenda-non-identity features add conflict indicators such as different identifiers and year conflict. The three relation heads are trained on the training split, and the merge policy exposes τ_w, τ_a, and τ_r as configurable thresholds. Unless a validation protocol overrides them, the implementation uses 0.5 for all three thresholds.

5.8. Feature and Head Specification

The implementation uses label provenance for supervision control, not as a predictive shortcut. Pair metadata such as label source, label strength, provenance, split, and hard-negative level are copied to prediction outputs for auditing, but they are excluded from the identity, agenda, and risk feature groups used by the relation heads. Table 7 gives the implementation-level contract used by the reported transformer variants.

Table 7: Feature and head specification for IAD-Risk transformer variants. Metadata fields are retained for auditing but are not used as model features.

Head or stage	Main input fields	Supervision or calculation	Output role
Identity head	Transformer distances, title similarity, author overlap, first-author match, venue similarity, year match, DOI/arXiv/OpenAlex identifier agreement.	Same-work labels from valid identity evidence, with provenance-aware masking.	Estimates p_{work} as positive merge evidence.
Agenda head	Transformer cosine, title and venue similarity, OpenAlex-style topic overlap, reference Jaccard similarity, year-distance score.	Same-agenda proxy or silver evidence when the source supports agenda relatedness.	Estimates p_{agenda} as relatedness evidence, not direct merge evidence.
ANI risk head	Transformer distances, title and author signals, topic overlap, reference overlap, year conflict, identifier agreement, and different-identifier conflicts.	Agenda-non-identity hard negatives from valid hard-negative sources.	Estimates p_{ani} as direct blocking evidence.
Risk gate	p_{work}, p_{agenda}, p_{ani}, τ_w, τ_a, τ_r, and cannot-link flags.	Computes $p_{\text{risk}} = \max\{p_{\text{ani}}, p_{\text{agenda}}(1 - p_{\text{work}})\}$.	Emits merge, block, or defer decisions.
Audit metadata	Pair ID, source pair ID, label strength, label source, provenance, split, and hard-negative level.	Copied from pair records and result artifacts.	Supports artifact review; it is not a training feature.

6. Experiments

6.1. Dataset Construction and Evidence Strength

Open-v2 is built by converting public scholarly metadata into a unified pair schema. DeepMatcher/py_entitymatching records provide the same-work gold layer, while OpenAlex topics and OpenCitations-style citation signals provide agenda-level silver hard negatives. Each pair keeps its label strength, provenance, split, and hard-negative flag so that a gold identity result is never averaged with a silver stress-test result without being identified as such.

The benchmark therefore has two roles. The gold layer measures ordinary duplicate matching ability, and the silver hard-negative layer measures whether a system over-merges topically close but distinct papers. This separation is important for interpreting the results: a representation baseline can look useful for scientific retrieval while still being unsafe for automatic work identity merging.

6.2. Split and Leakage Controls

The evaluation protocol treats split assignment as part of the evidence contract rather than as a preprocessing detail. Each pair carries a split field, label strength, label source, relation label, provenance, and hard-negative level. Training uses only the declared training split, threshold selection uses validation evidence when available, and held-out metrics are reported only after the operating point has been fixed. Metadata fields that identify source, provenance, or split are retained for auditing but are excluded from model features.

Table 8: Split and leakage controls used to interpret IAD-Bench results. The controls define what the current evidence supports and what stronger follow-up evidence would require.

Control	Manuscript role	Stronger evidence boundary
Train/dev/test split field	Separates fitting, threshold selection, and held-out reporting.	Full numerical reproduction requires released prediction files, configs, and split summaries.
Unordered pair leakage guard	Detects whether the same document pair appears across multiple splits.	A blocked leakage audit prevents held-out claims until duplicate pair assignments are removed.
Label-stratum coverage audit	Checks whether gold, proxy, and silver evidence appear in the intended evaluation strata.	Missing strata must be reported as evidence gaps rather than averaged into a single score.
Source-heldout readiness audit	Tests whether relation labels have enough independent label sources for source-heldout claims.	Broad source generalization remains conditional until source-heldout coverage is defensible.
Topic-heldout readiness audit	Tests whether enough topics exist to reserve unseen-topic evaluation slices.	Cross-topic stability should not be claimed when topic coverage is insufficient.

These controls reduce two common reviewer concerns. First, the model cannot gain accuracy from explicit provenance or split identifiers because those fields are audit metadata rather than predictive features. Second, a held-out result is interpreted only within the split and coverage regime that generated it. If a split-readiness audit blocks source-heldout or topic-heldout claims, the manuscript keeps the conclusion at the level of random-split or mechanism evidence instead of presenting it as broad generalization.

6.3. Experimental Questions

The experiments ask four questions. RQ1 tests whether IAD-Risk preserves same-work matching performance on gold labels. RQ2 tests whether it reduces false merges on agenda-level hard negatives. RQ3 examines whether the observed behavior is consistent with the proposed risk mechanism. RQ4 tests whether results remain interpretable under gold, proxy, and silver label strata.

Table 9: Evaluation protocol. Each question is tied to a label stratum and a metric so that evidence strength is not mixed across sources.

RQ	Evidence layer	Metric	Interpretation
RQ1	Gold identity pairs	Same-work F1	Duplicate matching ability
RQ2	Silver hard negatives	HNFMR	False-merge safety
RQ3	Mechanism analysis	FMR and HNFMR	Role of risk signals
RQ4	Gold/proxy/silver strata	Split metrics	Provenance-aware robustness

6.4. Evaluation Metrics

Same-work F1 is computed on pairs whose identity labels come from the gold layer. Ordinary false-merge rate (FMR) is the fraction of non-identity pairs that the system marks for automatic merging. Hard-negative false-merge rate (HNFMR) is the same error measured only on agenda-level hard negatives, where topic or citation evidence is high but identity evidence is negative. Reporting both FMR and HNFMR separates general matching errors from the specific identity-agenda confusion that this paper studies.

6.5. Threshold Selection and Uncertainty Reporting

Threshold-sensitive systems are evaluated under a fixed selection protocol rather than by choosing the best test threshold after observing results. Merge thresholds are selected on validation evidence when validation rows are available; otherwise, the default implementation threshold is reported as a fixed operating point. The test split is then used only for final metric reporting. This protocol is necessary because a small threshold change can alter FMR and HNFMR more strongly than it alters same-work F1.

Table 10: Threshold and uncertainty reporting protocol. The protocol separates what is required for the current main table from what is required for stronger follow-up claims.

Item	Reporting rule	Reviewer-facing purpose
Merge thresholds	Record τ_w, τ_a, τ_r, validation source, and fixed operating point.	Prevents post-hoc threshold selection on the test set.
Metric uncertainty	Report bootstrap intervals when prediction files and resampling logs are present in the artifact package.	Distinguishes stable effects from small numerical differences.
Risk metrics	Always report FMR and HNFMR with same-work F1.	Prevents a high F1 score from hiding unsafe hard-negative merges.
Ablation claims	Claim component causality only when no-risk-gate, no-ANI, and single-space variants are present.	Separates mechanism-consistent evidence from complete causal ablation evidence.
Artifact audit	Tie each numerical table to predictions, configs, logs, checksums, and commit identifiers.	Makes result-level reproduction possible without committing raw data.

6.6. Operating Point Disclosure

The main result table reports fixed operating points rather than post-hoc best test thresholds. Table 11 summarizes how each row family obtains its decision field and operating point. Exact threshold values, selected validation sources, and command logs must be verified from the released artifact manifest before using the table for numerical reproduction.

Table 11: Operating point disclosure for the Open-v2 result table. The table records how merge decisions are produced and what artifact is required for threshold audit.

Row family	Decision field	Operating point source	Audit requirement
Representation cosine baselines	Cosine score threshold on scientific document embeddings.	Fixed baseline metric summary; threshold chosen before final test reporting.	Score file, metric summary, and threshold entry.
RoBERTa pair classifier	Pair match probability threshold.	Fixed pair-classifier metric summary under the same pair schema.	Prediction file, metric summary, and model log.
IAD-Risk transformer variants	p_{work}, p_{ani}, and p_{risk} under the risk gate.	Model configuration; default $\tau_w = \tau_a = \tau_r = 0.5$ unless the artifact config records an override.	Prediction file, model JSON, thresholds, and checksums.

6.7. Threshold Sensitivity Evidence Status

Threshold stability is treated as an audit requirement, not as an unsupported robustness claim. The current Open-v2 table reports fixed operating points and therefore supports the mechanism interpretation under those operating points. A stronger statement that the method is stable across thresholds would require a threshold grid computed from the same prediction files, predefined threshold ranges, metric summaries for each setting, and checksums in the released artifact manifest.

Table 12: Threshold sensitivity evidence status. The table separates the current fixed-threshold claim from the additional artifact evidence required for threshold-stability claims.

Audit item	Current manuscript status	Required artifact before stronger claim
Fixed operating point	Reported through the operating-point disclosure table and tied to validation or default thresholds.	Configuration file recording τ_w, τ_a, τ_r, baseline thresholds, and selection source.
Threshold grid	The threshold grid is not reported as primary evidence in the current manuscript.	Sensitivity table generated from the same prediction files over predefined threshold ranges.
Metric stability	Interpreted only for the reported operating points.	Per-threshold F1, FMR, HNFMR, pair counts, random seeds, and command logs.
Artifact manifest	Required for numerical audit but not included in Git.	Manifest and checksums that bind prediction files, summaries, thresholds, and code commit.
Reviewer interpretation	Supports fixed-threshold false-merge control, not threshold-stable ranking across all operating points.	Stronger threshold-stability wording only after the released artifact contains the grid evidence.

6.8. Baselines

The main table reports two scientific representation cosine baselines and one supervised pair-classifier baseline: SciNCL cosine, SPECTER2 adapter cosine, and RoBERTa pair classification. These rows are evaluated through the same pair schema so that same-work F1 and hard-negative false-merge rate can be interpreted without mixing label strengths. Other repository baseline utilities are not used as primary manuscript evidence unless their metric summaries, prediction files, thresholds, and checksums are included in the released artifact package.

6.9. Implementation and Reproducibility

All reported systems are evaluated through the same pair-level schema, split field, and metric implementation. Threshold-sensitive baselines select merge thresholds on validation evidence and are then evaluated on held-out evidence. The main metrics are same-work F1 on gold identity pairs, ordinary false-merge rate, and HNFMR on agenda-level hard negatives. Rows with different pair counts in Table 15 should therefore be read as evidence for the same mechanism under different available result scopes, not as a single leaderboard.

The repository separates executable code from large data artifacts: raw third-party data remain outside Git, while processing scripts, fixture tests, schema contracts, and artifact-release rules remain version controlled. Reproduction starts from public sources, rebuilds IAD-Bench pair files through the documented CLI, and checks generated outputs with manifests and checksums. This organization is intended to make the main evidence reproducible without redistributing restricted or large source files. The supplementary material states the claim-evidence boundary for each major conclusion so that code-level reproducibility, artifact-level reproducibility, and manual-validation requirements are not conflated.

6.10. Artifact Reproduction Protocol

Because the repository does not redistribute large third-party data, reproducibility is divided into code-level reproduction and result-level artifact reproduction. Table 13 defines what each level proves. The key principle is that small fixtures verify the processing contracts, while the full reported tables require either independently obtained public source files or a released artifact package with manifests and checksums.

Table 13: Reproduction levels for auditing the repository and the reported evidence.

Level	Required input	Audit target
L0 code check	Source tree, package environment, and manuscript scripts	Verify that the CLI, tests, public-release scan, and LaTeX build run in a clean environment.
L1 fixture rebuild	Public miniature fixtures under `tests/fixtures`	Verify that DeepMatcher, SciRepEval-style, OpenAlex-style, and IAD-Bench adapters produce the documented pair schema.
L2 public-source rebuild	Independently obtained public raw files from the cited sources	Rebuild derived IAD-Bench files with the documented data-processing commands and inspect label provenance.
L3 result audit	Released tables, predictions, logs, manifests, and checksums	Verify the numerical results reported in the manuscript against immutable artifact files.

6.11. Claim-Evidence Boundary for Result Interpretation

The result interpretation follows a conservative claim-evidence boundary. Each reported conclusion must be tied to a label stratum, an output artifact, and a statement of what the evidence does not prove. This boundary is especially important because the benchmark combines gold identity pairs with proxy and silver stress-test evidence rather than treating all rows as one homogeneous test set.

Table 14: Claim-evidence boundary used to interpret the reported results. Each claim class is paired with the supporting evidence and the claims that are intentionally not made.

Claim class	Required support	Boundary not claimed
Identity-agenda confusion is a concrete false-merge pathway.	Hard-negative false-merge rate on agenda-level silver pairs, compared with single-space scientific representation baselines.	A prevalence estimate for all scholarly corpora.
IAD-Risk reduces risky automatic merges in the reported setting.	Held-out Open-v2 IAD-Risk predictions, thresholds, metric summaries, and checksums.	A broad method-ranking claim or a claim that every source distribution is covered.
IAD-Bench supports provenance-aware evaluation.	Pair schema fields for relation type, label source, label strength, provenance, split, and hard-negative level.	A claim that proxy or silver rows are human-gold labels.
Repository-level reproduction is possible without committing raw data.	Public code, fixture tests, documented data-processing commands, manifest rules, and PDF build scripts.	Full numerical reproduction without public source files or a released artifact package.

6.12. Result Audit Trail

Table 15 should be read together with the artifact boundary in Table 14. Each numerical row must be traceable to three audit objects: a prediction or score file, a metric summary produced from that file, and a checksum or manifest entry that fixes the artifact version. The Git repository verifies the schema, fixture rebuild, public-release boundary, manuscript build, and submission package, but it deliberately does not store raw third-party data, large model outputs, or full prediction files. Therefore the manuscript treats the Open-v2 table as a supported evidence snapshot and reserves full numerical reproduction for the released L3 artifact package.

6.13. Main Open-v2 Results

Table 15: Open-v2 evidence snapshot. Lower false-merge rate is better. Rows are reported only when supported by current output summaries; different pair counts indicate different available result scopes.

System	Pairs	Same-work F1 ↑	FMR ↓	HNFMR ↓
SciNCL cosine	10415	0.054	0.785	0.790
SPECTER2 adapter cosine	10415	0.044	0.999	0.999
RoBERTa pair classifier	10415	0.825	0.001	0.0001
IAD-Risk transformer (SciNCL)	1042 test	0.980	0.001	0.000
IAD-Risk transformer (SPECTER2)	1042 test	0.980	0.001	0.000

The Open-v2 evidence shows why identity-agenda disentanglement is necessary. Single-space scientific representation baselines preserve topical relatedness, but their merge threshold can produce high false-merge rates on OpenAlex silver hard negatives. IAD-Risk transformer variants keep high same-work F1 on the held-out Open-v2 test split while suppressing hard-negative false merges. RoBERTa pair classification is a strong baseline and narrows the gap substantially; therefore the paper does not make broad superiority claims. Instead, the main result is that relation heads plus an explicit risk score provide a transparent and controllable mechanism for false-merge suppression.

6.14. Scope Compatibility of the Open-v2 Table

The Open-v2 table is a scope-bounded evidence table, not a single leaderboard. Full pair-scope representation baselines test whether single-space scientific relatedness can over-merge agenda-level hard negatives across the available Open-v2 pairs. Held-out IAD-Risk rows test whether risk-gated transformer variants preserve same-work performance and suppress hard-negative merges under a split-based model-evaluation protocol. The valid comparison is therefore mechanistic: semantic relatedness alone is unsafe as automatic merge evidence, while explicit risk gating can block hard-negative merges in the reported held-out setting.

Table 16: Scope compatibility for interpreting the Open-v2 evidence snapshot. The table states what each row family supports and which stronger comparison remains outside the current evidence.

Row family	Pair scope	Supported interpretation	Not supported
Representation baselines	Full available Open-v2 pair scope.	Tests whether one-space scientific relatedness creates hard-negative merge risk.	A tuned neural pair-classifier ranking under the same held-out prediction file.
RoBERTa pair classifier	Full available Open-v2 pair scope.	Shows that a strong supervised pair classifier can suppress most hard-negative merges.	A component-level explanation of identity-agenda risk control.
IAD-Risk transformer variants	Held-out Open-v2 test scope.	Tests whether explicit identity, agenda, and ANI heads support risk-gated held-out decisions.	A general ranking over every baseline before all rows share the same released prediction scope.
Future stronger comparison	Same released prediction scope plus manual-validation slice.	Would support a stricter leaderboard and stronger label-precision claims.	A claim that this stronger comparison has already been completed by the current evidence snapshot.

This scope boundary is important for review. The evidence is sufficient for the paper's conservative claim that identity-agenda disentanglement is a practical false-merge control mechanism. It is not sufficient for a broad ranking claim until all row families are regenerated with the same released prediction scope, threshold-selection logs, checksums, and an independent manual-validation slice.

6.15. Extended Protocols and Boundaries

Open-v3 and source-heldout protocols are retained as follow-up evaluation paths, not as additional result evidence in the current manuscript package. These protocols are useful because they specify how source-diverse and domain-diverse validation should be audited before broader claims are made.

The manuscript therefore treats Open-v2 as the core mechanism demonstration and reserves Open-v3/source-heldout conclusions for a released artifact package that contains matched prediction scopes, threshold logs, checksums, and manual-validation evidence.

7. Mechanism and Error Analysis

The central mechanism question is whether false-merge control follows from explicit agenda-non-identity supervision and the derived risk score. The supported Open-v2 comparison gives mechanism-consistent evidence: single-space scientific representations can over-merge agenda-related hard negatives, while IAD-Risk makes agenda-non-identity evidence a direct blocking signal. This does not by itself establish a complete causal ablation. A complete ablation package must compare the full gate with variants that remove risk gating, remove agenda-non-identity supervision, or collapse identity and agenda evidence into one score.

Table 17: Mechanism evidence and reviewer-facing interpretation. The table separates current evidence from ablation outputs that would be required for a stronger causal claim.

Mechanism question	Current evidence	Reviewer-facing boundary
Does topical relatedness create merge risk?	Representation baselines show high false-merge rates on OpenAlex silver hard negatives.	Supports a targeted failure mode, not a universal prevalence estimate.
Does explicit risk gating suppress hard-negative merges?	IAD-Risk transformer variants keep low HNFMR under the reported Open-v2 split.	Supports the reported contract and thresholds; transfer must be rechecked on new source mixes.
Is the gain caused by each IAD-Risk component?	The current manuscript reports mechanism-consistent HNFMR evidence.	A full causal claim requires ablations removing risk gating, ANI supervision, and single-space separation.
Can cluster-level contamination be eliminated?	The method supports cannot-link-aware constrained union-find.	Guarantees require sufficient cannot-link coverage and cluster-level artifact audits.

The qualitative error taxonomy in Table 18 states which failure cases the framework is designed to expose. The taxonomy is diagnostic rather than a measured error distribution unless a released artifact contains per-category annotations and adjudication logs. This distinction matters because some categories, such as version boundaries, require human judgment beyond metadata-derived silver evidence.

Table 18: Error taxonomy for identity-agenda confusion. The table identifies the expected failure mode, the IAD-Risk control, and the audit evidence needed for stronger claims.

Error category	Typical source of confusion	IAD-Risk control	Stronger audit evidence
Same task, different contribution	Papers share a benchmark, model family, or problem statement.	Agenda-non-identity head and HNFMR reporting.	Human-reviewed hard-negative slice with pair notes.
Citation-neighborhood neighbor	Papers share references or citation community but are distinct works.	Reference-overlap features contribute to agenda evidence, not direct merge evidence.	Citation-derived pair provenance and checksum-fixed predictions.
Version or extension boundary	Preprint, short paper, conference, and journal records are similar but may not be mergeable.	Conservative defer/block behavior when identity evidence is incomplete.	Manual adjudication with version-policy labels.
Identifier conflict	DOI, arXiv, or OpenAlex identifiers disagree despite text similarity.	Different-identifier conflicts enter the ANI risk head and cannot-link logic.	Identifier-resolution logs and conflict audit table.
Sparse metadata	Abstract, venue, author, or identifier fields are missing or noisy.	Merge requires positive identity evidence rather than semantic similarity alone.	Missing-field stratified metrics and per-source diagnostics.

The error analysis emphasizes claim discipline. The current evidence supports identity-agenda risk modeling, stratified benchmark construction, and targeted false-merge suppression. It does not support claims that OpenAlex labels are human gold, that the system has broad domain superiority, that component-level ablations have already closed every mechanism question, or that manual annotation has already been completed.

8. Limitations

This study has four limitations. First, OpenAlex/OpenCitations hard negatives are silver labels, not human gold. They are useful for stress testing but must be reported separately from DeepMatcher gold labels. Second, SciRepEval/SciDocs-style proximity is a proxy for agenda relatedness, not proof of identity or non-identity. Third, source-heldout generalization remains mixed across model variants and should be expanded before making broad claims across scholarly domains. Fourth, an independently reviewed 500–1,000 pair validation slice would provide stronger evidence about label precision and difficult boundary cases; the supplementary material specifies the sampling, blinding, adjudication, and agreement-reporting protocol required before making stronger human-validation claims.

9. Threats to Validity

Construct validity depends on whether the label strata measure the intended relations. DeepMatcher pairs provide same-work identity evidence, but OpenAlex/OpenCitations hard negatives are silver stress-test pairs rather than manual non-identity gold. Internal validity depends on threshold selection and split separation; therefore threshold-sensitive systems must select thresholds on validation data and report held-out metrics. External validity is limited by the current source mix: the Open-v2 evidence supports the identity-agenda mechanism, while broader claims require more source-heldout and domain-diverse evaluation. Reproducibility depends on public data availability and stable processing scripts because the repository deliberately avoids redistributing large third-party data files.

Table 19 summarizes the main validity risks and the safeguards used in the manuscript. The table is intentionally conservative: a mitigation reduces a specific reviewer concern, but it does not turn proxy or silver evidence into human-adjudicated truth.

Table 19: Threats to validity and claim boundaries. Each risk is paired with the current mitigation and the remaining statement that should not be over-claimed.

Validity dimension	Main risk	Current mitigation	Remaining boundary
Construct validity	Silver hard negatives may contain metadata noise.	Label strength is preserved; silver rows remain stress-test evidence.	Not a replacement for manual non-identity gold.
Internal validity	Threshold choice or split leakage can inflate metrics.	Operating points, split fields, and pair-leakage audits are required.	Numerical claims need released predictions and logs.
External validity	Open-v2 may not cover every source, topic, or version policy.	Claims are tied to Open-v2 and source-heldout evidence boundaries.	Broad deployment requires domain-diverse validation.
Conclusion validity	HNFMR evidence may be mistaken for full causality.	Ablation requirements precede component-level causal claims.	Complete causality requires no-gate, no-ANI, and single-space variants.
Reproducibility	Raw data and large predictions are not stored in Git.	Public commands, fixtures, manifests, checksums, and artifact rules are provided.	Full numeric audit requires L2/L3 artifacts.
Operational validity	Pair errors can propagate through transitive merges.	Risk gating and cannot-link-aware clustering are described.	Cluster guarantees need complete cannot-link coverage.

10. Conclusion

IAD-Risk addresses a specific failure mode in scholarly entity matching: the tendency to confuse agenda similarity with work identity. By separating identity and agenda evidence and gating automatic merges through false-merge risk, the framework makes deduplication safer under stratified gold, proxy, and silver evaluation. The current evidence supports targeted false-merge suppression and a reproducible benchmark contract. Additional validation should expand independent manual review and source-heldout evaluation before the framework is used for broad method ranking across scholarly domains.

Data and Code Availability

The source code, benchmark construction scripts, public-release checks, and manuscript build scripts are organized in the project repository. After installation, the reproducibility entry points include CLI commands for DeepMatcher conversion, proximity conversion in the SciRepEval style, weak-label construction from OpenAlex and OpenCitations, IAD-Bench assembly, stratification checks, source-bias diagnostics, baseline execution, IAD-Risk training, bootstrap analysis, and manuscript PDF generation.

Raw third-party data are not redistributed in the repository because the original sources have their own access conditions and licenses. The repository instead provides data-processing commands, schema contracts, fixture tests, and artifact-release guidance so that independent readers can reproduce the benchmark construction from public sources and verify generated artifacts through manifests, checksums, command logs, and fixed commit identifiers.

Ethics Statement

This study uses public scholarly metadata and benchmark-style record linkage data. It does not involve human participants, clinical records, private user behavior, or animal experiments. Any future manual validation should be limited to scholarly record relation labels and should avoid collecting sensitive personal information.

Competing Interests

The authors declare no competing interests.

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